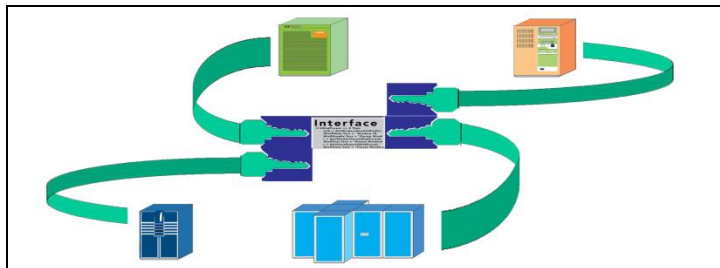
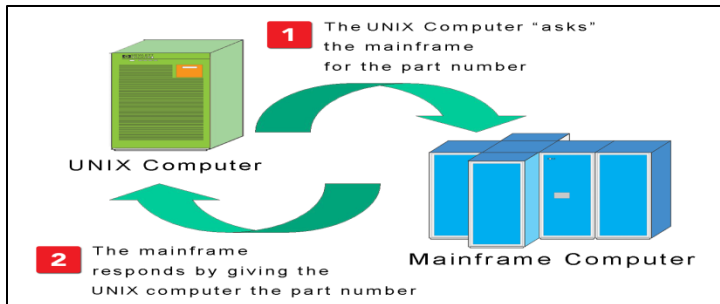
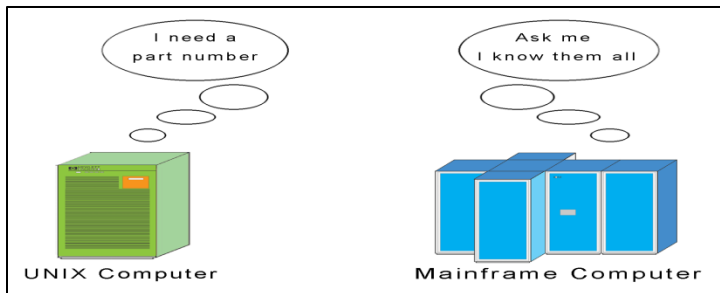


SaaS and SOA

Scenario

- XYZ's Clothing has been in business for over 5 years, and they've grown to the point where they can no longer operate efficiently as a single business unit. So XYZ splits the company into 4 subsidiaries.
 - XYZ's Clothing Group Of Companies (Head office)
 - XYZ's Shirts
 - XYZ's Pants
 - XYZ's Shoes

What is SOA?



- Data and functionality typically reside on more than one system (and application)

Applications need to be able to “talk to each other”

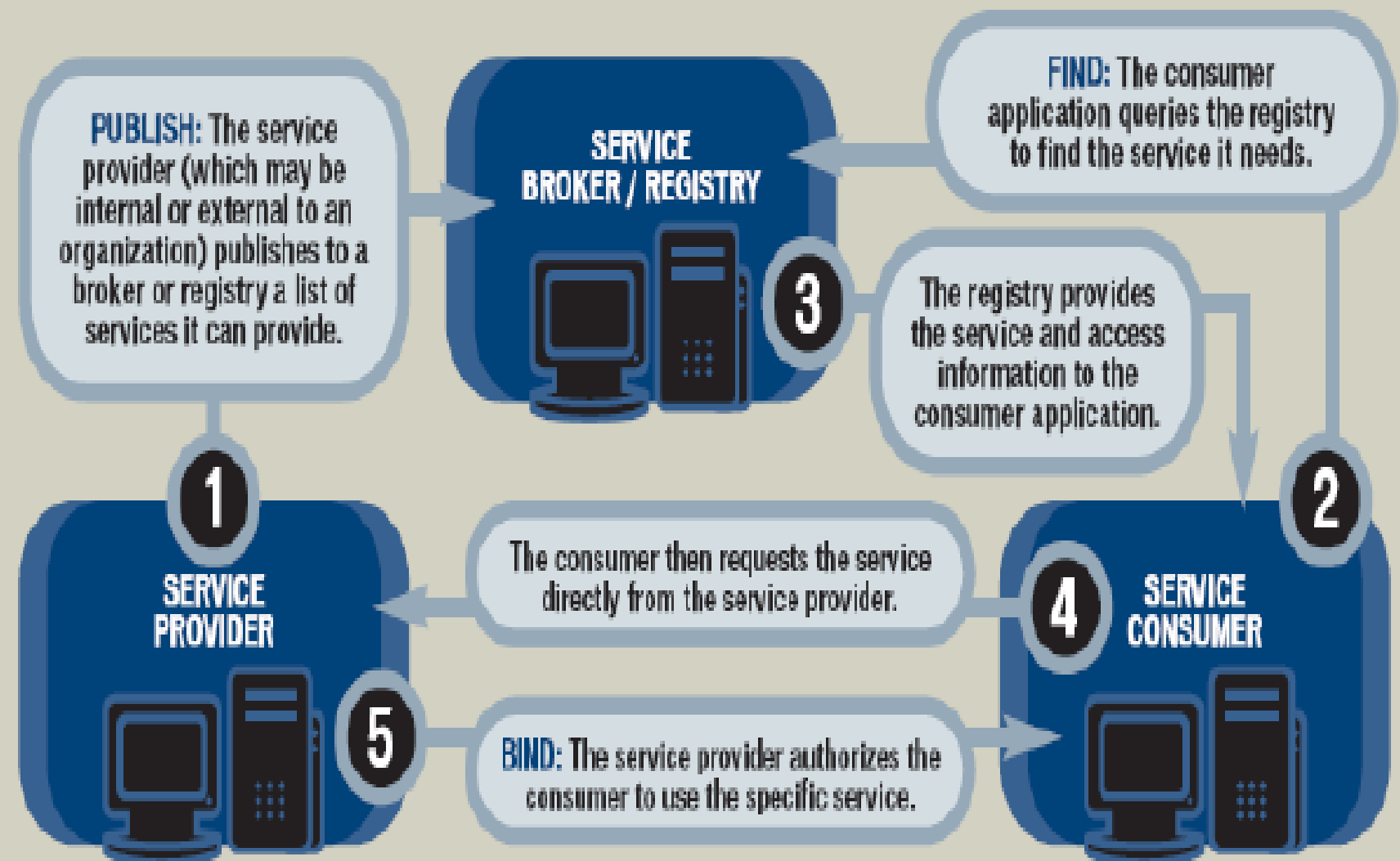
Proprietary or custom communication interfaces between applications

Definition

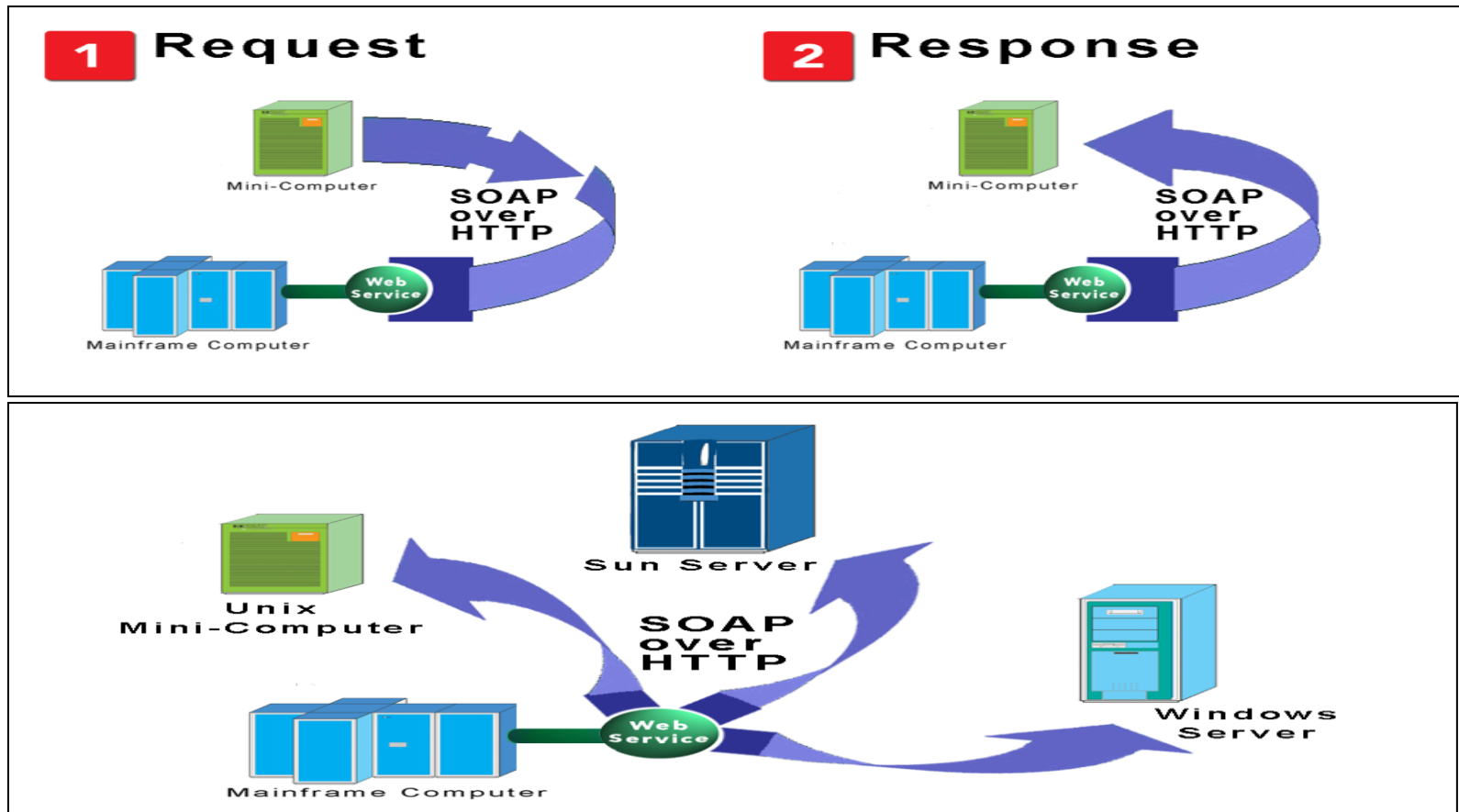
- IT architecture composed of software that has been exposed as “Services” – i.e. invoked on demand using a standard communication protocol.
- “Web Services” – software available as a “service” using Internet protocols.
- One software application talking to another using a standards-based (i.e. non-proprietary) language over a standards-based communication protocol.
- Universal “Dial Tone” between software applications
- An IT architecture that enables “loose coupling” of applications

How SOA Works

A Web services SOA involves three primary entities that communicate with one another: a producer or provider of services, a consumer or requester of services, and a directory, registry or broker of available services.



Request & Response



Core of SOA

- **XML** – Extensible Markup Language
- **SOAP** – Simple Object Access Protocol
- **WSDL** – Web Services Description Language
- **UDDI** - Universal Description, Discovery and Integration
- **ESB** – Enterprise Service Bus
- Key Concepts
 - Network Transparency
 - Virtualized endpoint
 - Self-describing software
 - Universally discoverable software
 - Universally understood software
 - Machine to machine interaction

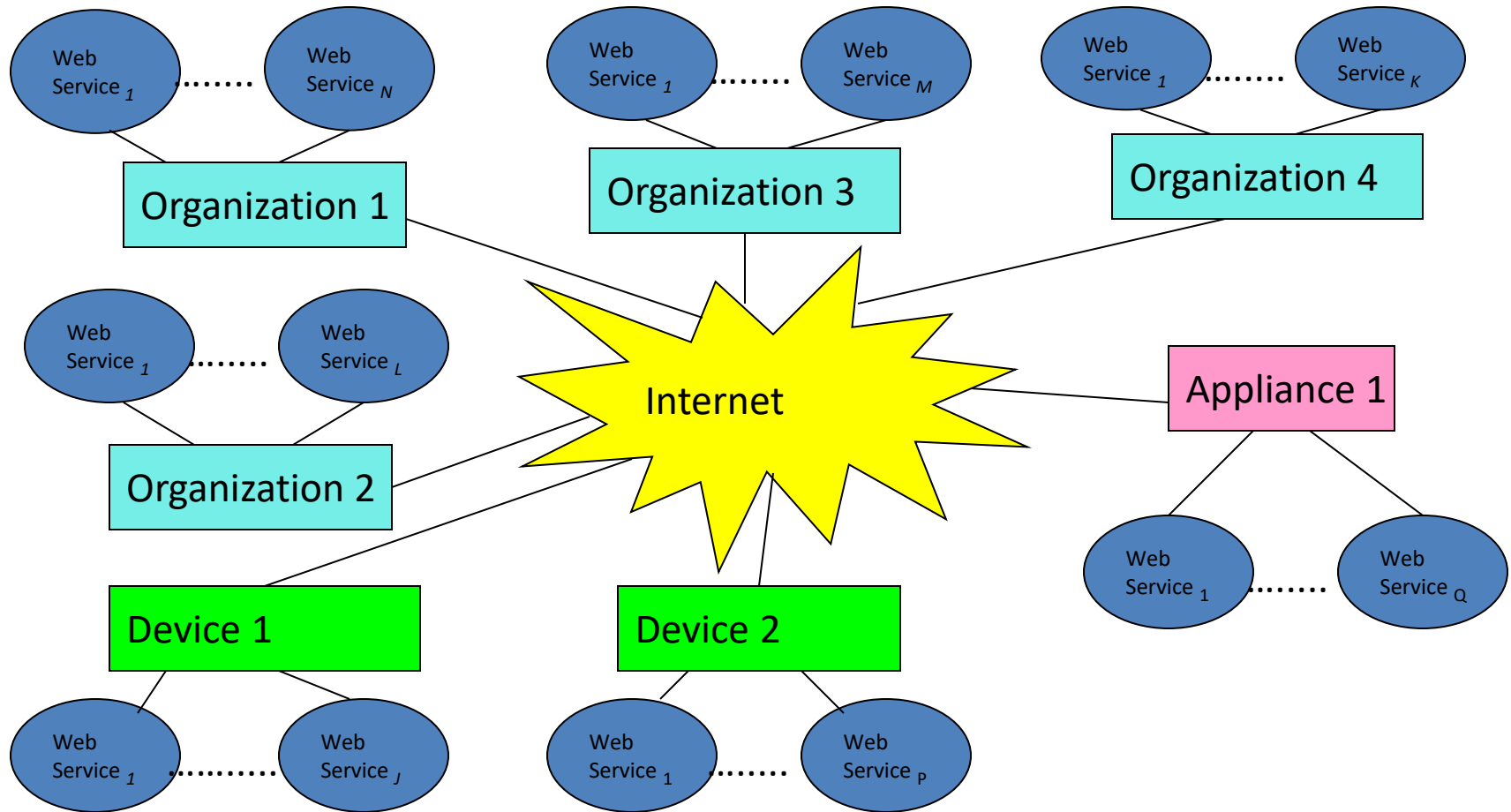
Features

- Means of designing and building software
- No reference to whether or not these services are deployed over the web
- Could deliver it via the Internet (don't have any effect if it is SOA or not)

SOA Usage & Supporting Platforms

- SOA Usage
 - B2B
 - Enterprise Application Integration (EAI)
 - Application to Application
 - Government
- Major Players in SOA Space
 - IBM: WebSphere SOA Product Suite
 - BEA: Aqualogic (WebLogic)
 - Oracle: Fusion Middleware
 - Microsoft: .NET
 - SAP: NetWeaver

Sample Web Service Topology



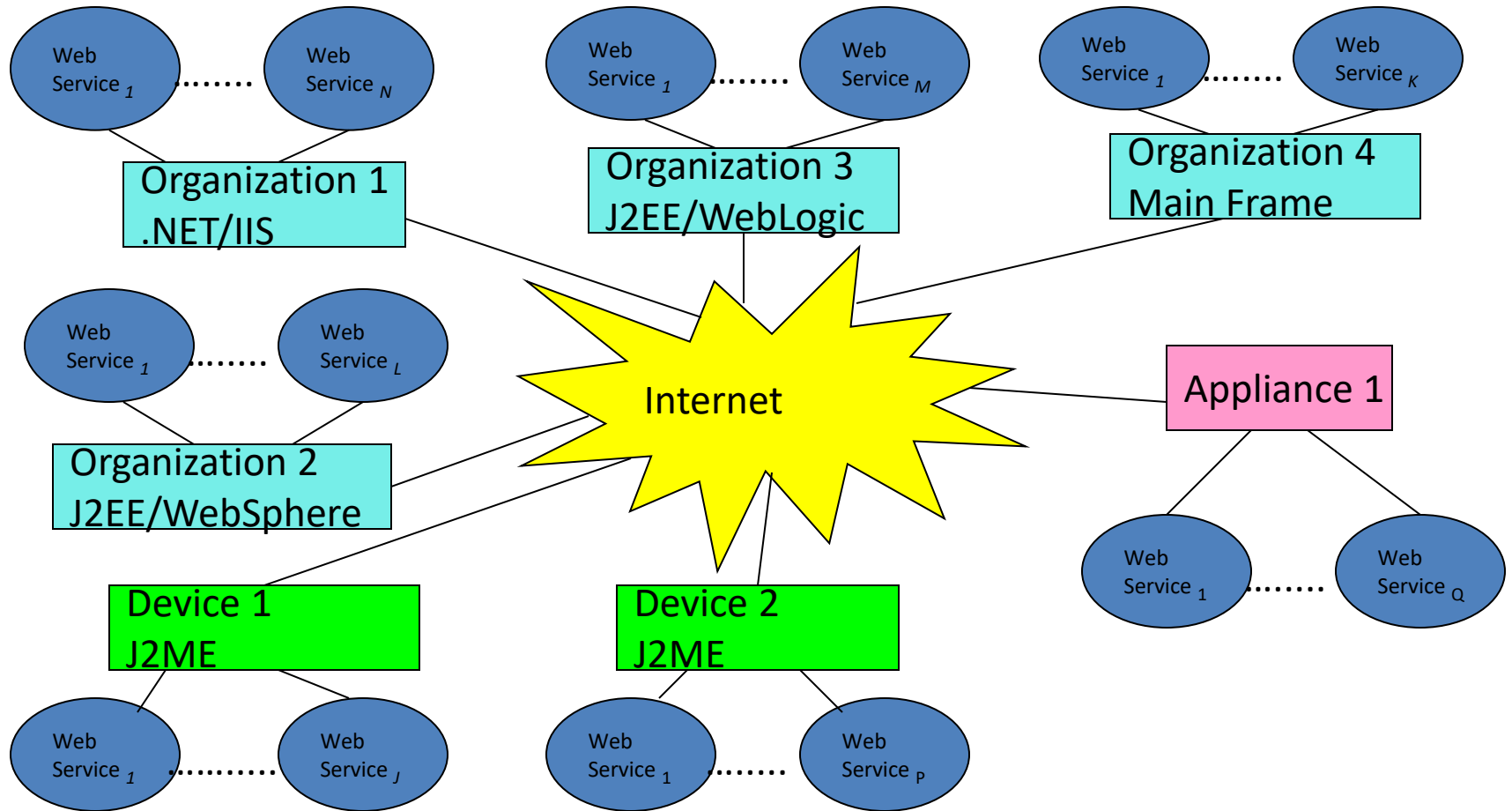
Challenges faced in SOA

- How do you develop Web Services in an organized and predictable way?
 - Where is a Web Service in its life cycle?
 - Concept? Development? QA? Testing? Deployed?
- Questions if you have a complex *ecosystem* of services
 - How do you manage them operationally?
 - What services are up/down, for how long, etc.
 - Are the services load balanced?
 - What are policies for accessing the endpoints?
 - How about security?

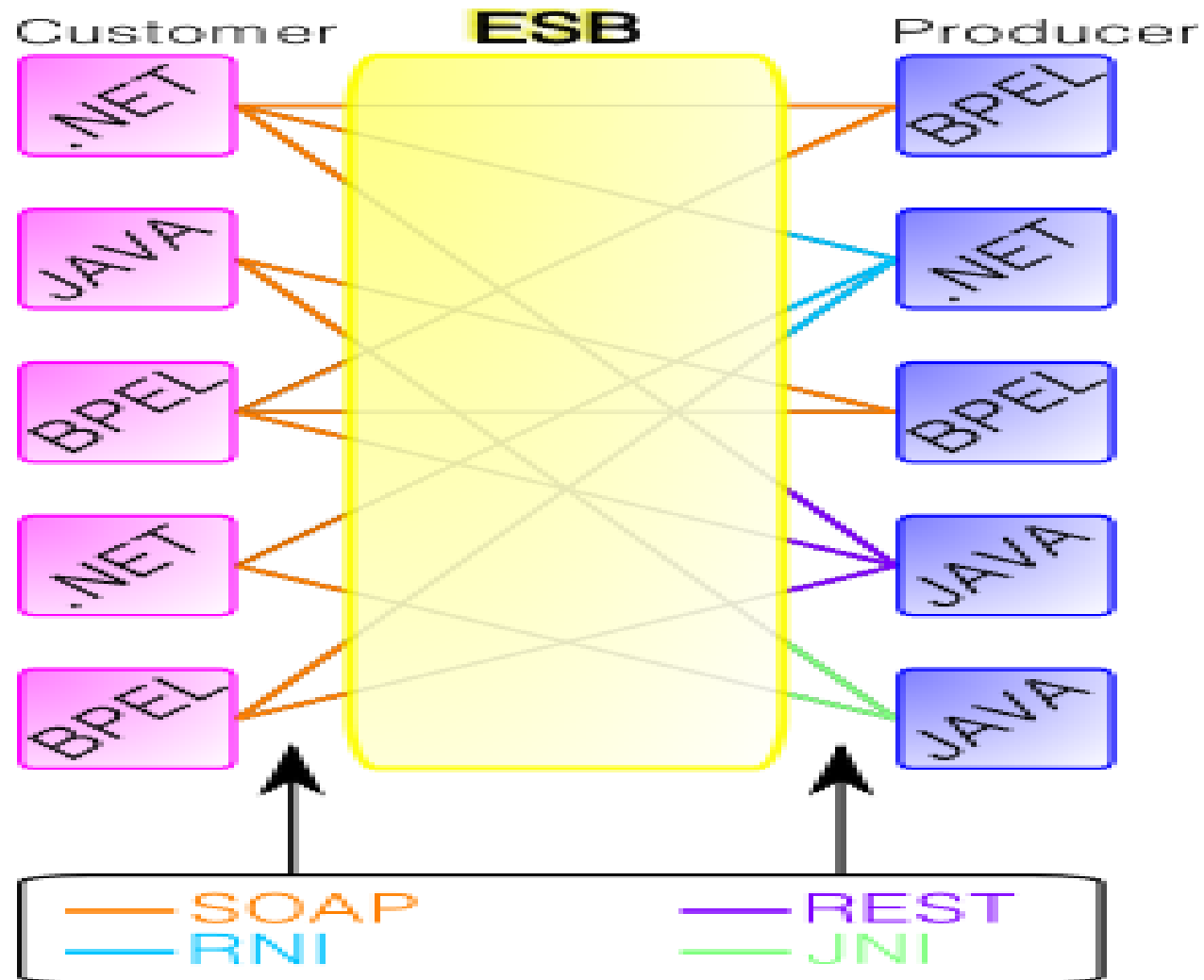
Steps

- Key Task: Identification & Analysis
 - Define Services
 - Define Policy
 - Define Profiles
 - Define Process
 - Define Test Cases
 - Information Architecture
 - Identify other assets

Platform Gov



ESB



REST and SOAP

- REST (Representational state transfer)-
 - an architectural style of the web services
 - defines a set of constraints to be used for creating Web services
 - work as a channel of communication between different computers or systems on the internet
 - Deliver fast performance and reliability

REST and SOAP

- SOAP(Simple Object Access Protocol)-
 - standard communication protocol
 - permits processes using different os to communicate via HTTP and Its XML
 - designed to create, recover, update and delete records like accounts, passwords, leads, and custom objects
 - already operating the all operating systems
 - Easily manipulate web services and get responses without caring about language and platforms

SaaS

Common types

- Desktops
- Google Apps
- Salesforce (mainly CRM utilities)
- Others(A2Zaps, OpenID)

Software-as-a-Service (SaaS)

- A software delivery model providing access to applications through the Internet as a Web-based service
- Frees users from complex hardware & software management by offloading to third parties, who build applications accessible through a Web browser
- Customers neither need install anything on their premises nor have to pay considerable up-front costs to purchase the software and the required licenses
- Users access the application website, enter their credentials and billing details, and can instantly use the application (customizable)

Features

- Application or service, deployed from a centralized datacenter across a network(Internet, Intranet, LAN, or VPN)
- Providing access and use on a recurring fee basis
- Users “rent,” “subscribe to,” “are assigned,” or “are granted access to” the applications from a central provider
- Business models vary according to the level
- Works for lower price and increased efficiency, or value-addition through customization

Core characteristics of SaaS

- Product sold to customer is application access
- Application is centrally managed
- Service delivered is one-to-many
- Service is an integrated solution delivered on the contract

SOAP vs. REST

#	SOAP	REST
1	A XML-based message protocol	An architectural style protocol
2	Uses WSDL for communication between consumer and provider	Uses XML or JSON to send and receive data
3	Invokes services by calling RPC method	Simply calls services via URL path
4	Does not return human readable result	Result is readable which is just plain XML or JSON
5	Transfer is over HTTP. Also uses other protocols such as SMTP, FTP, etc.	Transfer is over HTTP only
6	JavaScript can call SOAP, but it is difficult to implement	Easy to call from JavaScript
7	Performance is not great compared to REST	Performance is much better compared to SOAP - less CPU intensive, leaner code etc.

CRM and ERP Applications

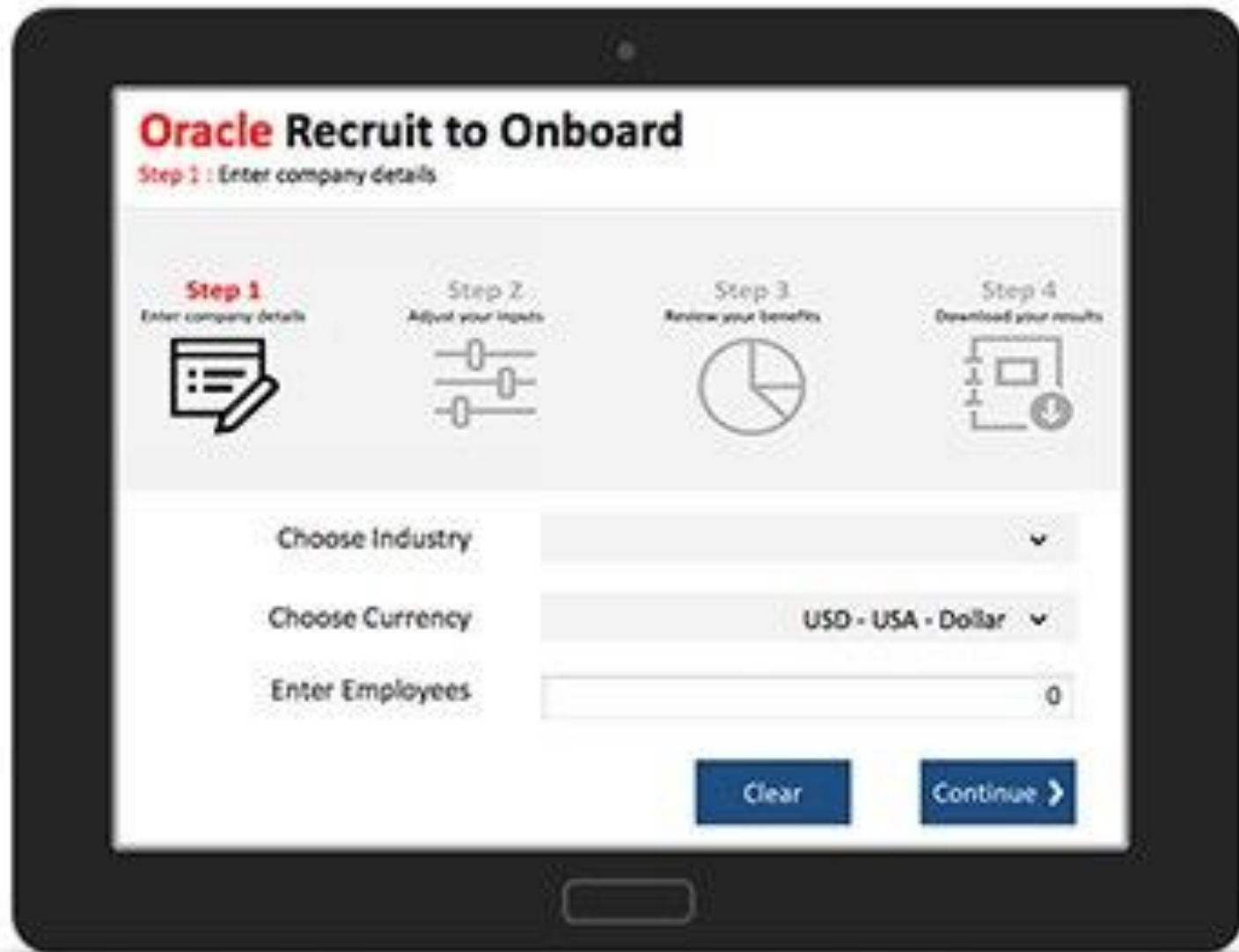
- Constitute common basic features in small to medium-sized and large enterprises
- Different needs can be satisfied with further customization

Example

Oracle Taleo Cloud Service

- Oracle's fully integrated talent acquisition solution
- Includes recruitment marketing and employee referrals to source talent
- Provide end-to-end recruiting automation
- Streamline employee onboarding

Interaction



The image shows a tablet displaying the Oracle Recruit to Onboard application. The interface is designed for a four-step process to onboard a company. Step 1, 'Enter company details', is the active step. It features a progress bar at the top with four steps: Step 1 (active), Step 2 (Adjust your inputs), Step 3 (Review your benefits), and Step 4 (Download your results). Below the progress bar, there are three input fields: 'Choose Industry' with a dropdown arrow, 'Choose Currency' with a dropdown menu showing 'USD - USA - Dollar', and 'Enter Employees' with a numeric input field showing '0'. At the bottom right, there are two buttons: 'Clear' and 'Continue' with a right-pointing arrow.

Oracle Recruit to Onboard
Step 1 : Enter company details

Step 1
Enter company details

Step 2
Adjust your inputs

Step 3
Review your benefits

Step 4
Download your results

Choose Industry

Choose Currency

Enter Employees

Clear

Continue >

Advantages

- Application is shared across multiple users
- adapted to specific needs with little further customization